

Elliptic Equilibrium and Boundary-Corrected Transseries for Hyperbolic Fekete Designs

An explicit Fabius design limit, screened crystallization,
and a finite-size determinant transition

Research manuscript prepared for Vladimir Reshetnikov

Abstract

The common-digit Fabius construction in the ProveIt repository leads to the covariance kernel $\text{sech}(u - v)$ and asks for the limiting distribution of parameters maximizing its determinant on a fixed interval. We solve the equilibrium-measure part of that question explicitly. The density is an elliptic deformation of the arcsine law, its constant potential is a ratio of complete elliptic integrals, and its large-interval expansion admits a convergent exponential–logarithmic representation with a certified inverse. These continuum results are connected with classical Green capacity rather than presented as a new discovery of elliptic capacity.

The finite-design analysis goes further. We prove explicit two-sided energy bounds, a dimension-independent bound on deviations from equal spacing when the mean gap stays positive, and a dimension-uniform dilute boundary correction. The latter quantifies the failure of the equally spaced design: its energy exceeds the optimum by $2(n-3)e^{-6h}/(n-1) + O(e^{-8h})$, with an absolute remainder uniform in the number of nodes. It yields a deterministic transition $\det R_n^* \rightarrow \exp(-4e^{-2s})$ at $h = \frac{1}{2} \log n + s$, together with three finite-size correction orders. Exact q -product factorizations and a modular bulk expansion provide the transseries connection. Full proofs, reproducible numerical diagnostics, source provenance, and further research questions are included. Microscopic fixed-interval endpoint spacings and a complete joint MacMahon asymptotic are explicitly left open.

Status. A mathematical research manuscript with proofs and numerical checks, not a Lean formalization or an independently refereed paper. Claims of worldwide priority are not made. The main new-development target is the uniform finite-design boundary analysis, not the classical elliptic formulas.

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1 Problem, provenance, and scope

1.1 The precise repository question

The starting point is the repository report *Common-Digit Fabius Zonoids: Exact volumes, hyperbolic-secant geometry, Bernoulli Gaussianization, and parameter jets* [1]. Its section on hyperbolic node design proves the existence and uniqueness of a configuration maximizing

$$\prod_{1 \leq i < j \leq n} \tanh(u_j - u_i), \quad 0 \leq u_1 < \dots < u_n \leq L. \quad (1)$$

The later question labeled `question:design-limit` asks for the limiting empirical measure, an explicit constant-potential density, and endpoint spacings. The question labeled `question:q-barnes` asks for joint asymptotics and modular transseries of the equally spaced product.

The source was inspected at the repository commit

0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

The relevant file has Git blob identifier `dabfc501dfe952d826001703a087b47424568581`. Its exact path and the inspected source ranges are recorded in [section A](#) and in the archive's `PROVENANCE.md`. The present paper does not infer that a statement in that research report is machine-checked. The covariance and determinant facts needed here are rederived below.

1.2 Normalization

Throughout, logarithms are natural. Set

$$V(x) = -\log \tanh x \quad (x > 0), \quad V(0) = +\infty, \quad E_n(u) = \sum_{i < j} V(u_j - u_i). \quad (2)$$

The minimum on the interval is denoted by $E_n^*(L)$. We use

$$P_n^*(L) = e^{-E_n^*(L)}, \quad \mathcal{D}_n(L) = e^{-2E_n^*(L)}. \quad (3)$$

Thus P_n^* is the optimal product in (1), while $\mathcal{D}_n$ is the optimal covariance determinant. This factor of two is kept explicit. For a probability measure μ on $[0, L]$, define

$$\mathcal{E}_L(\mu) = \iint V(|u - v|) d\mu(u) d\mu(v). \quad (4)$$

The notation $K(k)$ always uses the *elliptic modulus*:

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}. \quad (5)$$

Software interfaces using the parameter $m = k^2$ must be supplied with k^2 , not with k .

1.3 What is established here

The continuum question is answered by

$$\rho_L(u) = \frac{2}{K(k) \sqrt{(1 - e^{-4u})(1 - e^{-4(L-u)})}}, \quad k = \sqrt{1 - e^{-4L}}. \quad (6)$$

Its equilibrium energy is

$$I_L = \pi \frac{K(e^{-2L})}{K(\sqrt{1 - e^{-4L}})}. \quad (7)$$

Every sequence of optimal empirical measures converges to $\rho_L(u)du$ for fixed L , and $\log \mathcal{D}_n(L)/(n(n-1)) \rightarrow -I_L$.

For the discrete problem put $m = n - 1$, $h = L/m$, and $q = e^{-2h}$. With

$$U_n(h) = \sum_{k=1}^{n-1} (n-k)V(kh), \quad (8)$$

we prove the dimension-uniform formula

$$E_n^*((n-1)h) = U_n(h) - 2\frac{n-3}{n-1}q^3 + O(q^4), \quad n \geq 3. \quad (9)$$

The constant in $O(q^4)$ is independent of n . Consequently, at $h = \frac{1}{2} \log n + s$,

$$\mathcal{D}_n((n-1)h) \rightarrow \exp(-4e^{-2s}). \quad (10)$$

A three-order correction to the energy is proved in [theorem 9.2](#). The uniformity in (9), rather than a fixed- n Taylor series, is what permits this growing-dimensional substitution.

1.4 Classical ingredients and novelty boundary

Green equilibrium, transfinite diameter, and convergence of minimizing configurations are classical themes; Saff's account [2] provides background. The elliptic modulus of a disk slit by a geodesic segment is also classical. In particular, the capacity discussion in Nasser–Rainio–Vuorinen [3] identifies the corresponding Grötzsch elliptic ratio. We give an elementary derivation in the precise coordinates of the repository problem. The elliptic and eta expansions used later are standard formulas recorded by NIST [4, 5].

The contributions developed specifically in this manuscript are the quantitative finite-design estimates, the bounded total gap-defect theorem, and especially the absolute, dimension-uniform boundary correction and its finite-size transition consequences. A targeted literature check did not establish a published match for that particular package of statements. That is not a proof of priority; the theorems are offered with proofs, not with an unsupported claim to settle a famous general Fekete problem.

2 From common digits to a strictly convex design problem

2.1 A direct covariance calculation

Let $U_0, U_1, \dots$ be independent uniform random variables on $[0, 1]$, and for $0 \leq r < 1$ set

$$X_r = (1-r) \sum_{j \geq 0} r^j U_j. \quad (11)$$

The series converges absolutely for every sample. Independence gives

$$\text{Var}(X_r) = \frac{1-r}{12(1+r)}, \quad (12)$$

$$\text{Cov}(X_r, X_t) = \frac{(1-r)(1-t)}{12(1-rt)}. \quad (13)$$

After standardization and the substitutions $r = \tanh u$, $t = \tanh v$,

$$\text{Corr}(X_{\tanh u}, X_{\tanh v}) = \frac{\sqrt{(1-r^2)(1-t^2)}}{1-rt} = \text{sech}(u-v). \quad (14)$$

This is the common-digit covariance connection in the repository report. It is an exact identity, not a Gaussian approximation.

Proposition 2.1 (Cauchy determinant identity). *For distinct real $u_1, \dots, u_n$, the matrix $R(u) = [\operatorname{sech}(u_i - u_j)]_{i,j=1}^n$ satisfies*

$$\det R(u) = \prod_{i < j} \tanh^2(u_j - u_i) = e^{-2E_n(u)}. \quad (15)$$

Proof. Put $t_i = e^{2u_i}$. Then $R_{ij} = 2\sqrt{t_i t_j} / (t_i + t_j)$. The Cauchy determinant formula gives

$$\det \left[\frac{1}{t_i + t_j} \right] = \frac{\prod_{i < j} (t_j - t_i)^2}{\prod_{i,j} (t_i + t_j)}.$$

Multiplication by the row and column factors contributes $\prod_i 2t_i$, which cancels the diagonal factors in the denominator. The remaining factor for each pair is $((t_j - t_i)/(t_j + t_i))^2$, as required. $\square$

2.2 The interaction and the optimizer

For $x > 0$, elementary differentiation and the power series for the logarithm give

$$V(x) = 2 \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{e^{-2rx}}{r}, \quad (16)$$

$$V'(x) = -\frac{2}{\sinh(2x)}, \quad V''(x) = \frac{4 \cosh(2x)}{\sinh^2(2x)} > 0. \quad (17)$$

Thus V is positive, decreasing, strictly convex, logarithmically singular at zero, and exponentially decreasing at infinity.

Proposition 2.2 (Unique finite optimizer). *For $n \geq 2$ and $L > 0$, the minimizing configuration exists and is unique among ordered configurations. It has $u_1 = 0$, $u_n = L$, and $u_i + u_{n+1-i} = L$. Its interior nodes satisfy*

$$\sum_{j < i} V'(u_i - u_j) - \sum_{j > i} V'(u_j - u_i) = 0, \quad 2 \leq i \leq n-1. \quad (18)$$

Proof. Extend the energy by $+\infty$ at collisions. Lower semicontinuity on the compact ordered simplex gives a minimum; a strictly ordered test configuration has finite energy, so a minimizer has no collision. The leftmost point must be zero: moving it to zero strictly increases all its distances if it was positive and hence decreases the energy. The same argument applies to the rightmost point.

Write the positive gaps as $d_i = u_{i+1} - u_i$, $1 \leq i \leq m = n-1$, with $\sum_i d_i = L$. Each summand of the energy is a convex function of a sum of consecutive gaps. The nearest-neighbor contribution $\sum_i V(d_i)$ is strictly convex. Hence the whole energy is strictly convex on this affine simplex. This proves uniqueness. Reflection preserves the objective, so uniqueness implies the symmetry. Differentiating at the interior minimizer gives (18). $\square$

For $n = 2, 3$ the equally spaced design is optimal. For larger n it is not generally optimal, despite becoming an accurate approximation in dilute regimes. The size and structure of that difference are quantified below.

3 Exact elliptic equilibrium on a fixed interval

3.1 A useful change of variables

Let

$$b = e^{2L}, \quad k = \sqrt{1 - b^{-2}}, \quad k' = b^{-1}. \quad (19)$$

Under $t = e^{2u}$ and $s = e^{2v}$,

$$V(|u - v|) = \log \left| \frac{t + s}{t - s} \right|, \quad 1 \leq t, s \leq b. \quad (20)$$

The candidate density in these variables is

$$w_b(t) = \frac{b}{K(k)\sqrt{(t^2 - 1)(b^2 - t^2)}}, \quad 1 < t < b. \quad (21)$$

Indeed,

$$\int_1^b \frac{dt}{\sqrt{(t^2 - 1)(b^2 - t^2)}} = \frac{K(k)}{b}. \quad (22)$$

For example, substitute $t^2 = 1 + (b^2 - 1) \sin^2 \theta$; the left side becomes $\int_0^{\pi/2} (1 + (b^2 - 1) \sin^2 \theta)^{-1/2} d\theta$. Changing θ to $\pi/2 - \theta$ factors out b^{-1} and gives (22).

Lemma 3.1 (An arcsine Cauchy integral). *For $A < B$,*

$$\int_A^B \frac{dy}{(y - z)\sqrt{(y - A)(B - y)}} = \frac{\pi}{\sqrt{(A - z)(B - z)}} \quad (z < A), \quad (23)$$

where the square roots on the right are positive. For $A < z < B$ the principal value of the same integral is zero.

Proof. The affine cosine substitution $y = (A + B)/2 + (B - A) \cos \theta/2$ reduces the first identity to the elementary integral $\int_0^\pi (c + d \cos \theta)^{-1} d\theta = \pi/\sqrt{c^2 - d^2}$ for $c > |d|$. It follows by $\tan(\theta/2)$ substitution. The same substitution in the principal-value case gives the principal value of $\int_0^\infty (\alpha - \beta x^2)^{-1} dx$ for $\alpha, \beta > 0$. Its logarithmic antiderivative has equal limits at zero and infinity, and the two logarithmic divergences at its pole cancel. Its principal value is therefore zero. $\square$

Theorem 3.2 (Explicit equilibrium measure and potential). *The density ρ_L in (6) is a probability density on $(0, L)$. Its potential is constant on the whole closed interval:*

$$\int_0^L V(|u - v|) \rho_L(v) dv = I_L = \pi \frac{K(k')}{K(k)}, \quad 0 \leq u \leq L. \quad (24)$$

It is the unique probability measure minimizing $\mathcal{E}_L$.

Proof. Normalization and change of density. Equation (22) normalizes (21). Multiplication by $dt/du = 2e^{2u}$ and factoring the two square roots gives exactly (6).

Constancy in the interior. Let

$$W(t) = \int_1^b \log \left| \frac{t + s}{t - s} \right| w_b(s) ds.$$

On every compact subinterval of $(1, b)$ the density is smooth, and logarithmic-potential differentiation is given by a principal value. This can also be proved directly by subtracting the density's value at $s = t$ in the local singular integral; the subtracted numerator is $O(s - t)$ and is integrable after differentiation. Thus

$$\begin{aligned} W'(t) &= -\frac{2b}{K(k)} \text{PV} \int_1^b \frac{s ds}{(t^2 - s^2)\sqrt{(s^2 - 1)(b^2 - s^2)}} \\ &= -\frac{b}{K(k)} \text{PV} \int_1^{b^2} \frac{dy}{(t^2 - y)\sqrt{(y - 1)(b^2 - y)}} = 0, \end{aligned}$$

by theorem 3.1. Consequently W is constant on $(1, b)$. The square-root edge behavior of w_b makes its logarithmic potential continuous up to both endpoints. For completeness, the contribution of

an endpoint interval of length ε is bounded by a constant multiple of $\sqrt{\varepsilon}(1 + |\log \varepsilon|)$; away from that interval ordinary dominated convergence applies.

Evaluation of the constant. For $0 \leq t < 1$ the logarithm is nonsingular and $W(0) = 0$. Differentiating and applying (23) yields

$$W'(t) = \frac{b\pi}{K(k)\sqrt{(1-t^2)(b^2-t^2)}}.$$

Integration from zero to one, followed by $t = \sin \theta$, gives $W(1) = \pi K(1/b)/K(k)$. Endpoint continuity identifies this with the constant on the support.

Minimality and uniqueness. Let μ denote the measure just constructed. If a competing probability measure ν has infinite energy, there is nothing to prove. Otherwise, constant potential gives $\iint V(|u-v|) d\mu(u) d\nu(v) = I_L$. For the signed measure $\sigma = \nu - \mu$, all required integrals are absolutely defined, since $|\sigma| \leq \mu + \nu$. Hence

$$\mathcal{E}_L(\nu) - I_L = \iint V(|u-v|) d\sigma(u) d\sigma(v). \quad (25)$$

Every kernel $e^{-a|u-v|}$, $a > 0$, is strictly positive definite on finite signed measures. In the Fourier convention $\hat{\sigma}(\xi) = \int e^{-i\xi u} d\sigma(u)$,

$$\iint e^{-a|u-v|} d\sigma(u) d\sigma(v) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2a}{a^2 + \xi^2} |\hat{\sigma}(\xi)|^2 d\xi \geq 0. \quad (26)$$

Fourier inversion is justified because $2a/(a^2 + \xi^2)$ is integrable. Equality implies $\hat{\sigma} = 0$ and therefore $\sigma = 0$. Expanding V by (16) in (25) is valid by absolute integrability against $|\sigma| \otimes |\sigma|$. Every term is nonnegative, and already the $r = 1$ term is strictly positive unless $\sigma = 0$. This proves the assertion. $\square$

Remark 3.3 (Relation to classical capacity). The map $r = \tanh u$ changes (20) into the unit-disk Green kernel $\log |(1-rt)/(r-t)|$ on a real segment. The classical Grötzsch expression for the same energy is

$$I_L = \frac{\pi}{2} \frac{K(\operatorname{sech} L)}{K(\tanh L)}. \quad (27)$$

The equivalence with (7) is an elliptic Landen transformation. With the convention that condenser capacity is 2π divided by the minimum of this Green energy, it is $2\pi/I_L$. The formulas in [3] supply the classical capacity interpretation. Our derivation of theorem 3.2 does not require the conformal mapping theorem or an imported formula for the equilibrium density.

4 Discrete convergence and the two continuum edge regimes

Theorem 4.1 (Convergence of optimal designs). *For fixed $L > 0$, let $u^{(n)}$ minimize E_n and set $\mu_n = n^{-1} \sum_{i=1}^n \delta_{u_i^{(n)}}$. Then*

$$\mu_n \Rightarrow \mu_L = \rho_L(u) du, \quad \frac{2E_n^*(L)}{n(n-1)} \longrightarrow I_L. \quad (28)$$

In particular,

$$(P_n^*(L))^{2/(n(n-1))} \longrightarrow e^{-I_L}, \quad \mathcal{D}_n(L)^{1/(n(n-1))} \longrightarrow e^{-I_L}. \quad (29)$$

Proof. Choose n independent points with distribution μ_L . Their expected energy is $n(n-1)I_L/2$. Some configuration has energy no larger than this expectation, so $2E_n^*(L)/(n(n-1)) \leq I_L$.

For the converse, put $V_M(x) = \min(V(x), M)$, with $V_M(0) = M$. It is bounded and continuous. Along any weakly convergent subsequence $\mu_n \Rightarrow \nu$,

$$\iint V_M(|u - v|) d\mu_n(u) d\mu_n(v) \leq \frac{2E_n^*(L)}{n^2} + \frac{M}{n}.$$

Weak convergence of product measures gives the corresponding limiting inequality for ν . Letting $M \rightarrow \infty$ and using monotone convergence,

$$I_L \leq \mathcal{E}_L(\nu) \leq \liminf_n \frac{2E_n^*(L)}{n^2} \leq I_L.$$

Uniqueness in [theorem 3.2](#) forces $\nu = \mu_L$. Compactness of probability measures on $[0, L]$ now shows that the full sequence converges. The product and determinant statements follow from [\(3\)](#). $\square$

4.1 What the density says at an endpoint

Proposition 4.2 (Continuum edge and quantile scales). *For fixed $L > 0$,*

$$\rho_L(u) = \frac{1 + O(u)}{kK(k)\sqrt{u}}, \quad u \downarrow 0, \quad (30)$$

and the right endpoint has the reflected expansion. If $Q_L(p)$ is the quantile of μ_L , then

$$Q_L(p) = \frac{(kK(k))^2}{4} p^2 (1 + O(p^2)), \quad p \downarrow 0. \quad (31)$$

Proof. Expand $1 - e^{-4u} = 4u(1 + O(u))$ and $1 - e^{-4(L-u)} = k^2(1 + O(u))$ in [\(6\)](#). Integration gives $\mu_L([0, u]) = 2\sqrt{u}/(kK(k))(1 + O(u))$. Inverting this expression yields [\(31\)](#). $\square$

Important distinction. Formula [\(31\)](#) is a continuum statement. Weak convergence in [theorem 4.1](#) does not justify inserting $p = 1/n$ to obtain the first discrete gap. Such a claim would require a microscopic theorem not proved here. This is precisely the part of the repository's endpoint-spacing question that remains open.

4.2 Short intervals and screened long intervals

Proposition 4.3 (Arcsine limit, bulk flattening, and surface excess). *As $L \downarrow 0$, the rescaled measures converge in total variation to the arcsine law:*

$$L\rho_L(Lx) \longrightarrow \frac{1}{\pi\sqrt{x(1-x)}} \quad \text{in } L^1(0, 1). \quad (32)$$

As $L \rightarrow \infty$, put $\ell = L + \log 2$. For fixed $u > 0$,

$$\ell\rho_L(u) \longrightarrow (1 - e^{-4u})^{-1/2}. \quad (33)$$

Uniformly where u and $L - u$ both tend to infinity, $\ell\rho_L(u) \rightarrow 1$. The single-edge excess of the limiting boundary layer is

$$\int_0^\infty \left((1 - e^{-4u})^{-1/2} - 1 \right) du = \frac{\log 2}{2}. \quad (34)$$

Proof. For the short-interval limit, $k \rightarrow 0$ and $K(k) \rightarrow \pi/2$. The two factors in the square root of [\(6\)](#) are $4Lx(1 + O(L))$ and $4L(1 - x)(1 + O(L))$, uniformly for $0 < x < 1$ in the relative sense. The rescaled densities are bounded by a fixed multiple of $[x(1 - x)]^{-1/2}$, so dominated convergence proves [\(32\)](#).

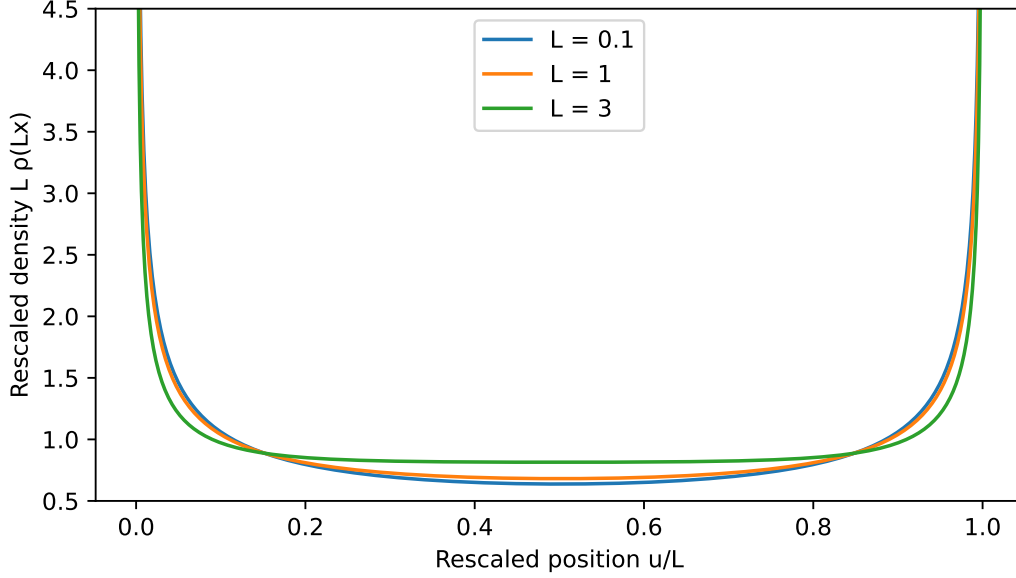


Figure 1: The exact rescaled equilibrium density. Short intervals approach the arcsine profile; longer intervals have a flatter interior and thinner boundary layers. The vertical axis is truncated only for legibility.

For the long-interval limit, the elliptic expansion proved in the next section gives $K(k) = 2\ell + O(Le^{-4L})$. Substitute this into the exact density to obtain (33) and the bulk statement. Finally, $y = e^{-4u}$ transforms (34) into

$$\frac{1}{4} \int_0^1 \frac{(1-y)^{-1/2} - 1}{y} dy.$$

With $t = \sqrt{1-y}$ this equals $\frac{1}{2} \int_0^1 (1+t)^{-1} dt = \frac{1}{2} \log 2$. $\square$

The effective length $L + \log 2$ therefore has a geometric interpretation: the two edges each supply excess mass corresponding to half of $\log 2$. The arcsine edge singularity survives while the interior becomes nearly flat. This makes precise the qualitative prediction in the repository.

5 Convergent capacity transseries and certified inversion

5.1 A convergent exponential–logarithmic normal form

Let $z = e^{-4L}$, $H = \log(4e^{2L}) = 2L + 2\log 2$, and define

$$a_j = \left(\frac{(1/2)_j}{j!} \right)^2, \quad d_j = 2(H_{2j} - H_j), \quad d_0 = 0, \quad (35)$$

$$A(z) = \sum_{j \geq 0} a_j z^j, \quad D(z) = \sum_{j \geq 1} a_j d_j z^j, \quad B(z) = D(z)/A(z). \quad (36)$$

Here $H_j = \sum_{r=1}^j 1/r$ is a harmonic number, with $H_0 = 0$. The convergent complete-elliptic-integral expansions in NIST [4, §19.12] give

$$K(\sqrt{z}) = \frac{\pi}{2} A(z), \quad K(\sqrt{1-z}) = H A(z) - D(z). \quad (37)$$

The $\log 4$ in H absorbs the $j = 0$ digamma constant in the usual formula.

Theorem 5.1 (Capacity expansion with a positive error certificate). *For every $L > 0$,*

$$I_L = \frac{\pi^2}{2(H - B(z))} = \frac{\pi^2}{4(L + \log 2 - B(e^{-4L})/2)}. \quad (38)$$

The Taylor series of B converges for $|z| < 1$ and begins

$$B(z) = \frac{z}{4} + \frac{13z^2}{128} + \frac{23z^3}{384} + \frac{2701z^4}{65536} + \frac{5057z^5}{163840} + O(z^6). \quad (39)$$

Let A_N, D_N be the truncations through degree N , and put

$$I_{L,N} = \frac{\pi^2}{2(H - D_N(z)/A_N(z))}. \quad (40)$$

Then

$$0 \leq I_L - I_{L,N} \leq \frac{\pi^2 \log 2}{4L^2} \frac{z^{N+1}}{1-z}. \quad (41)$$

Proof. The normal form follows by dividing the two expressions in (37). The functions A, D are analytic on $|z| < 1$. Moreover,

$$A(z) = \frac{2}{\pi} \int_0^{\pi/2} (1 - z \sin^2 \theta)^{-1/2} d\theta$$

has positive real part there: $1 - z \sin^2 \theta$ has positive real part, and so does its principal inverse square root. Thus A has no zero in the unit disk, proving analyticity of B . The coefficients in (39) follow by ordinary power-series division.

For real $0 < z < 1$, $0 < a_j \leq 1$, and d_j increases from zero to $2 \log 2$. Consequently D/A is a weighted average of the d_j , and adding its tail can only increase the average. In detail,

$$0 \leq B(z) - \frac{D_N(z)}{A_N(z)} \leq 2 \log 2 \sum_{j>N} a_j z^j \leq 2 \log 2 \frac{z^{N+1}}{1-z}.$$

Both $H - B$ and $H - D_N/A_N$ are at least $H - 2 \log 2 = 2L$. Subtracting the two reciprocals in (38) and (40) gives (41). $\square$

For example, with $\ell = L + \log 2$,

$$I_L = \frac{\pi^2}{4\ell} \left[1 + \frac{e^{-4L}}{8\ell} + \left(\frac{13}{256\ell} + \frac{1}{64\ell^2} \right) e^{-8L} + O\left(\frac{e^{-12L}}{\ell} \right) \right] \quad (42)$$

as $L \rightarrow \infty$. More generally, $(H - B)^{-1} = \sum_{r \geq 0} B^r / H^{r+1}$ is a convergent expansion in the analytic block $B(z)$ for every real $L > 0$, since $0 \leq B < 2 \log 2 < H$. Expanding the blocks into exponential sectors yields an absolutely convergent exponential-logarithmic series for sufficiently large L : then the absolutely summed Taylor coefficients of B are smaller than H . It is not necessary to invoke Borel summation or resurgence for this particular expansion.

5.2 Inverse interval design and exponentially invisible corrections

For a prescribed small equilibrium energy $I > 0$, one may ask for the interval length L producing it. This connects the equilibrium result directly with the repository's inversion program.

Theorem 5.2 (A certified inverse-capacity transseries). *The map $L \mapsto I_L$ is continuous and strictly decreasing from $+\infty$ to zero. Write its inverse as $L = L(I)$. Set*

$$X = \frac{\pi^2}{4I}, \quad Q = e^{-\pi^2/I}, \quad \varepsilon = 16Q. \quad (43)$$

For $16Q < 1/64$, there is an analytic function $w(\varepsilon)$ at zero such that

$$L(I) = X - \log 2 + w(\varepsilon), \quad w = \frac{1}{2}B(\varepsilon e^{-4w}). \quad (44)$$

Its coefficients have the nonrecursive formula

$$[\varepsilon^j]w(\varepsilon) = \frac{1}{2^j} [t^{j-1}]B'(t)e^{-2jB(t)}, \quad j \geq 1. \quad (45)$$

In particular,

$$L(I) = \frac{\pi^2}{4I} - \log 2 + 2Q - 3Q^2 + \frac{8}{3}Q^3 - \frac{3}{2}Q^4 + \frac{12}{5}Q^5 + O(Q^6). \quad (46)$$

The function w is analytic for $|\varepsilon| < 1/64$, and if w_N is its Taylor polynomial of degree N , then

$$|w(\varepsilon) - w_N(\varepsilon)| \leq \frac{1}{16} \frac{(64|\varepsilon|)^{N+1}}{1 - 64|\varepsilon|}. \quad (47)$$

Proof. Continuity follows from the elliptic formula. If $L_2 > L_1$, dilating μ_{L_1} to $[0, L_2]$ strictly decreases its energy because V is strictly decreasing and the measure is nonatomic. Minimality then gives $I_{L_2} < I_{L_1}$. The estimate $I_L \geq V(L)$ proves divergence at zero, and (38) gives the limit at infinity. Rearranging (38) gives $L = X - \log 2 + B(e^{-4L})/2$. Since $e^{-4(X - \log 2)} = 16e^{-\pi^2/I}$, this is (44).

Here is an explicit analytic existence estimate. On $|z| \leq r < 1/2$, $a_j \leq 1$, $d_j \leq 2 \log 2$ imply

$$|B(z)| \leq \frac{2 \log 2 r}{1 - 2r}, \quad |B'(z)| \leq \frac{2 \log 2}{(1 - 2r)^2}. \quad (48)$$

The second bound follows by differentiating D/A , bounding A', D' by geometric derivative sums, and using $|A| \geq (1 - 2r)/(1 - r)$. For $|\varepsilon| \leq 1/64$ and $|w| \leq 1/16$, one has $|\varepsilon e^{-4w}| \leq e^{1/4}/64 < 1/48$. The map on the right of (44) maps the closed $1/16$ disk strictly into itself by (48); its derivative in w has modulus at most $2(1/48) 2 \log 2 / (1 - 2/48)^2 < 1/2$. Uniform contraction therefore gives a unique solution, analytic in ε , with $|w| \leq 1/16$ throughout that parameter disk. Cauchy's coefficient estimate gives (47).

For the coefficient formula, put $t = \varepsilon e^{-4w}$, so that $t = \varepsilon e^{-2B(t)}$ and $w = B(t)/2$. The ordinary Lagrange–Bürmann formula gives (45); alternatively it follows by changing variables in the coefficient contour integral. Substitution of (39) yields (46). $\square$

All the corrections in (46) beyond $\pi^2/(4I) - \log 2$ are smaller than every positive power of I . The result is a convergent transseries example with a transparent analytic certificate, not merely a formal asymptotic inversion. Its underlying elliptic-nome inversion is classical; the formulation and error bound are adapted here to interval design.

6 Finite certificates and screened crystallization

The fixed- L limit describes a high-density regime. It must not be used without a new argument when L grows with the number of points. A direct finite inequality controls a complementary regime and will also supply a simple numerical certificate.

Theorem 6.1 (Explicit finite lower and upper bounds). *For every $n \geq 2$ and $L > 0$, put $h = L/(n - 1)$ and*

$$B_n(L) = \sum_{k=1}^{n-1} (n - k) V\left(\frac{kL}{n - k}\right). \quad (49)$$

Then

$$B_n(L) \leq E_n^*(L) \leq U_n(h). \quad (50)$$

Moreover,

$$0 \leq U_n(h) - B_n((n-1)h) \leq \mathcal{C}(h), \quad \mathcal{C}(h) = \sum_{k \geq 1} hk(k-1)(-V'(kh)) < \infty. \quad (51)$$

The function $\mathcal{C}$ is bounded on every compact subinterval of $(0, \infty)$.

Proof. For any gap vector $d_1, \dots, d_m > 0$, $m = n-1$, define the k -step distances $s_i^{(k)} = d_i + \dots + d_{i+k-1}$, $1 \leq i \leq n-k$. Every gap occurs in at most k of these distances, so $\sum_i s_i^{(k)} \leq kL$. Jensen's inequality and the fact that V is decreasing give

$$\sum_{i=1}^{n-k} V(s_i^{(k)}) \geq (n-k)V\left(\frac{\sum_i s_i^{(k)}}{n-k}\right) \geq (n-k)V\left(\frac{kL}{n-k}\right).$$

Summing over k proves the lower bound. The equal-gap configuration gives the upper bound.

For the uniform difference, write $x = kh$ and $y = k(n-1)h/(n-k)$. Since $y - x = kh(k-1)/(n-k)$ and V is convex,

$$(n-k)(V(x) - V(y)) \leq hk(k-1)(-V'(kh)).$$

Summing yields (51). Exponential decay of $-V'(kh)$ proves convergence and compact-uniform boundedness. $\square$

Corollary 6.2 (Determinant enclosure). *For all n, L as above,*

$$e^{-2U_n(h)} \leq \mathcal{D}_n(L) \leq e^{-2B_n(L)}. \quad (52)$$

More generally, any trial design with energy E_{trial} gives the lower determinant bound $e^{-2E_{\text{trial}}}$.

The inequalities are mathematical certificates. Floating-point evaluations of them, without directed rounding, are not certified interval arithmetic. That distinction is maintained in the numerical section.

6.1 A bound on the total gap defect

Define the convexity defect

$$D_h(x) = V(x) - V(h) - V'(h)(x-h), \quad x > 0. \quad (53)$$

It is nonnegative and vanishes only at $x = h$.

Theorem 6.3 (Bounded-defect crystallization). *For an optimal gap vector of mean h ,*

$$\sum_{i=1}^{n-1} D_h(d_i) \leq \mathcal{C}(h). \quad (54)$$

For every $0 < a < b < \infty$, there is a constant $C_{a,b}$ independent of n such that, whenever $a \leq h \leq b$,

$$\sum_{i=1}^{n-1} \min\{|d_i - h|^2, |d_i - h|\} \leq C_{a,b}. \quad (55)$$

In particular, for each fixed $\eta > 0$ the number of gaps with $|d_i - h| \geq \eta$ is bounded independently of n , and

$$\sum_{i=1}^{n-1} |d_i - h| = O_{a,b}(\sqrt{n}). \quad (56)$$

The empirical distribution of the rescaled nodes u_i/L converges to uniform measure on $[0, 1]$ in this growing-interval regime.

Proof. In the shell-by-shell proof of [theorem 6.1](#), every shell exceeds its lower bound by a nonnegative amount. The first shell exceeds its lower bound by

$$\sum_i V(d_i) - (n-1)V(h) = \sum_i D_h(d_i),$$

because $\sum_i (d_i - h) = 0$. Thus the left side of (54) is at most $E_n^* - B_n$, and this is at most $U_n - B_n \leq \mathcal{C}(h)$.

Uniformly for $h \in [a, b]$, strict positivity and continuity of V'' near h give $D_h(x) \geq c|x - h|^2$ when $|x - h|$ is small. As $x \rightarrow \infty$, $D_h(x)/x \rightarrow -V'(h)$ uniformly in $h \in [a, b]$, and $\min_{[a, b]}(-V') > 0$. As $x \downarrow 0$, $D_h(x) \rightarrow \infty$ uniformly in $h \in [a, b]$. On the remaining compact set the quotient by $\min(|x - h|^2, |x - h|)$ has a positive minimum. These facts prove (55).

Split the sum in (56) at $|d_i - h| = 1$. The large deviations have bounded total absolute size by (55); Cauchy–Schwarz bounds the sum of the small deviations by $O(\sqrt{n})$. Since $u_1 = 0$,

$$\max_i |u_i - (i-1)h| \leq \sum_j |d_j - h| = O(\sqrt{n}).$$

Division by $L = (n-1)h \geq (n-1)a$ gives a uniform $O(n^{-1/2})$ distance between the rescaled configuration and the uniform grid. Convergence of empirical measures follows. $\square$

The theorem does not assert that every gap tends to h . A bounded number of persistent boundary defects is entirely compatible with its conclusion. This is why the term “bounded-defect crystallization” is used rather than an unqualified uniform-spacing assertion.

6.2 Thermodynamic energy per point

Let

$$S(h) = \sum_{k \geq 1} V(kh), \quad T(h) = \sum_{k \geq 1} kV(kh). \quad (57)$$

Both sums converge for $h > 0$.

Corollary 6.4 (Bulk energy in the screened regime). *Uniformly for h in any compact subinterval of $(0, \infty)$,*

$$E_n^*((n-1)h) = nS(h) + O(1). \quad (58)$$

Consequently, if $h_n \rightarrow h > 0$,

$$\frac{E_n^*((n-1)h_n)}{n} \rightarrow S(h), \quad \mathcal{D}_n((n-1)h_n)^{1/n} \rightarrow e^{-2S(h)}. \quad (59)$$

Proof. The exact identity

$$U_n(h) = nS(h) - T(h) + R_n(h), \quad R_n(h) = \sum_{k \geq n} (k-n)V(kh) \geq 0 \quad (60)$$

shows $U_n = nS + O(1)$ compact-uniformly in h . Combine it with (51); continuity of S follows by uniform convergence on compact h -intervals. $\square$

7 Exact products and the modular bulk transseries

7.1 The finite product separated into bulk, surface, and tail

For $0 < q < 1$ define

$$(q; q)_\infty = \prod_{j \geq 1} (1 - q^j), \quad M(q) = \prod_{j \geq 1} (1 - q^j)^{-j}. \quad (61)$$

The latter is the MacMahon product. We use it as an explicit analytic product; no combinatorial enumeration theorem is needed.

Proposition 7.1 (Exact q -product factorization). *With $q = e^{-2h}$,*

$$S(h) = \log(q^2; q^2)_\infty - 2 \log(q; q)_\infty, \quad (62)$$

$$T(h) = 2 \log M(q) - \log M(q^2), \quad (63)$$

$$R_n(h) = 2 \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{q^{r(n+1)}}{r(1-q^r)^2}. \quad (64)$$

In particular, the equal-grid product factors as

$$e^{-U_n(h)} = \left(\frac{(q; q)_\infty^2}{(q^2; q^2)_\infty} \right)^n \frac{M(q)^2}{M(q^2)} e^{-R_n(h)}. \quad (65)$$

Proof. The identity $V(kh) = \log(1 - q^{2k}) - 2 \log(1 - q^k)$ gives (62) and (63) by summation. Insert (16) in (60). Every term is positive, so the rearrangement is justified, and

$$\sum_{k \geq n} (k - n) q^{rk} = q^{rn} \sum_{j \geq 0} j q^{rj} = \frac{q^{r(n+1)}}{(1 - q^r)^2}.$$

The factorization follows by exponentiating (60). $\square$

This identifies an exact MacMahon surface factor and an exact finite-size tail. It does *not* by itself furnish the complete joint $n \rightarrow \infty$, $h \downarrow 0$ asymptotic requested by the repository: $T(h)$ and $R_n(h)$ may both become large in that limit, and neither can be silently absorbed into an $O(1)$ remainder.

7.2 Two convergent exponential scales for the bulk

Define $\sigma_1(j) = \sum_{d|j} d$.

Theorem 7.2 (Direct and modular bulk expansions). *For $h > 0$, set $q = e^{-2h}$ and $Q = e^{-\pi^2/h}$. Then*

$$S(h) = \sum_{j \geq 1} a_j^{\text{bulk}} q^j, \quad a_j^{\text{bulk}} = 2 \sum_{\substack{r|j \\ r \text{ odd}}} \frac{1}{r}, \quad (66)$$

$$S(h) = \frac{\pi^2}{8h} + \frac{1}{2} \log \frac{h}{2\pi} + \sum_{j \geq 1} c_j Q^j, \quad c_j = \frac{-\sigma_1(j) + 4\mathbf{1}_{2|j}\sigma_1(j/2)}{j}. \quad (67)$$

Both series converge absolutely for all real $h > 0$. Their coefficients satisfy $|a_j^{\text{bulk}}| \leq 2j$ and $|c_j| \leq 2j$. Consequently either tail, after degree N and with its corresponding variable $x = q$ or $x = Q$, is bounded by

$$2 \frac{x^{N+1}((N+1) - Nx)}{(1-x)^2}. \quad (68)$$

The first modular terms are

$$-Q + \frac{1}{2}Q^2 - \frac{4}{3}Q^3 + \frac{5}{4}Q^4 - \frac{6}{5}Q^5 + \frac{2}{3}Q^6 + \dots. \quad (69)$$

Proof. Expanding (16) and collecting $j = rk$ gives (66). For the modular form, the product definition and the transformation $\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau)$ of Dedekind's eta function [5, §23.18] imply the real identity

$$\log(e^{-t}; e^{-t})_\infty = -\frac{\pi^2}{6t} + \frac{1}{2} \log \frac{2\pi}{t} + \frac{t}{24} + \log(e^{-4\pi^2/t}; e^{-4\pi^2/t})_\infty. \quad (70)$$

Apply it with $t = 4h$ and $t = 2h$ in (62). The linear terms in h cancel, leaving

$$S(h) = \frac{\pi^2}{8h} + \frac{1}{2} \log \frac{h}{2\pi} + \log(Q; Q)_\infty - 2 \log(Q^2; Q^2)_\infty. \quad (71)$$

The elementary coefficient identity $\log(x; x)_\infty = -\sum_{j \geq 1} \sigma_1(j) x^j / j$ now proves (67).

For the bounds, $a_j^{\text{bulk}} \leq 2j$ is immediate by bounding each reciprocal by one. Also $\sigma_1(j) \leq j^2$. For even j the second term in c_j has magnitude at most $4(j/2)^2/j = j$, so $|c_j| \leq 2j$; the odd case is smaller. Summing $2 \sum_{j > N} j x^j$ proves (68). $\square$

The two scales are complementary. For large h , e^{-2h} is the natural dilute parameter. For small h , the non-algebraic correction to the bulk is governed by $e^{-\pi^2/h}$. Both are convergent representations of the same analytic function; their appearance does not establish a Stokes phenomenon. A genuine divergent or resurgent analysis of the MacMahon surface term would be a different task.

8 The dimension-uniform dilute boundary correction

This section isolates the part of the analysis that cannot be obtained by simply evaluating the equally spaced product. All vector norms below are ordinary Euclidean or sup norms on the finite-dimensional gap space. The constants in the lemmas are absolute: they do not depend on its dimension.

Let $m = n - 1 \geq 2$ and write

$$d_i = h + \delta_i, \quad \sum_{i=1}^m \delta_i = 0, \quad q = e^{-2h}. \quad (72)$$

Let $P = I - m^{-1} \mathbf{1} \mathbf{1}^\top$ be the orthogonal projection onto the zero-sum subspace. Its Euclidean operator norm is one and its sup-norm operator norm is at most two. Set

$$e = (1, 0, \dots, 0, 1)^\top \in \mathbb{R}^m, \quad (73)$$

where for $m = 2$ both coordinates are one.

8.1 Boundary localization of the grid force

Let $g = \nabla_d E_n$ at the equal-gap vector $(h, \dots, h)$. Define

$$b(h) = \sum_{k \geq 1} k V'(kh). \quad (74)$$

This is the force each gap would see in a translation-invariant infinite chain; projection removes that common component.

Lemma 8.1 (Uniform grid-force estimate). *As $q \downarrow 0$, uniformly for all $m \geq 2$,*

$$Pg = 4q^2 Pe + O_{\ell^2}(q^3), \quad \|Pg\|_2 = O(q^2), \quad \|Pg\|_\infty = O(q^2). \quad (75)$$

More explicitly, for q in a fixed sufficiently small interval,

$$0 \leq g_i - b(h) \leq \frac{4}{(1 - q^2)(1 - q)^2} (q^{i+1} + q^{m-i+2}). \quad (76)$$

Proof. Let $N_{i,k}$ be the number of intervals of k consecutive gaps that contain gap i , with $N_{i,k} = 0$ if $k > m$. Then

$$g_i = \sum_{k \geq 1} N_{i,k} V'(kh), \quad 0 \leq k - N_{i,k} \leq (k - i)_+ + (k - (m - i + 1))_+.$$

For $k \leq m$, the last expression counts the intervals lost at the two ends; for $k > m$ it is at least k . Since $-V'(kh) = 4q^k/(1 - q^{2k}) \leq 4q^k/(1 - q^2)$, summing the two geometric arithmetic tails proves (76). Their Euclidean norms are $O(q^2)$, independent of m .

At $k = 1$ there is no boundary deficit. At $k = 2$ it is exactly the vector e , so this shell contributes $-V'(2h)e = 4q^2e + O_{\ell^2}(q^6)$ to $g - b(h)\mathbf{1}$. For every k , the deficit vector obeys $\|(k - N_{i,k})_{i=1}^m\|_2 \leq 2k^{3/2}$: when $k \leq m$, it is supported in two edge strips of length at most k , and when $k > m$ its norm is $k\sqrt{m} < k^{3/2}$. Thus the shells with $k \geq 3$ have total Euclidean norm at most $8(1 - q^2)^{-1} \sum_{k \geq 3} k^{3/2} q^k = O(q^3)$. Apply P to obtain (75). $\square$

8.2 A local optimizer with dimension-independent control

For δ in a fixed small sup-norm ball, (16) gives the absolutely convergent expansion

$$E_n(h + \delta) = 2 \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{1}{r} \sum_{k=1}^m q^{rk} \sum_{i=1}^{m-k+1} \exp \left(-2r \sum_{j=i}^{i+k-1} \delta_j \right). \quad (77)$$

Here $h + \delta$ abbreviates the gap vector, not a vector of node positions.

Lemma 8.2 (Uniform local existence and Hessian bounds). *There exist absolute $q_0 > 0$ and $C < \infty$ such that, for all $m \geq 2$ and $0 < q < q_0$, the global minimizing gap vector satisfies*

$$\|\delta\|_\infty \leq Cq, \quad \|\delta\|_2 \leq Cq. \quad (78)$$

Throughout the segment joining this vector to the grid,

$$\nabla_d^2 E_n = 8qI + O_{2 \rightarrow 2}(q^2). \quad (79)$$

All constants are independent of m .

Proof. First work in the zero-sum subspace with the sup norm and normalize the energy as $F_q = E_n/(2q)$. The $(r, k) = (1, 1)$ term of (77) is $\sum_i e^{-2\delta_i}$ and has Hessian $4 \text{diag}(e^{-2\delta_i})$. For any other pair (r, k) , the absolute row sum of its Hessian is bounded by

$$4rk^2 q^{rk-1} e^{2rk\|\delta\|_\infty}. \quad (80)$$

Indeed at most k windows contain a given gap, each window contributes k entries to that row, and differentiating twice contributes $4r^2$. For $\|\delta\|_\infty \leq C_0 q$ and q sufficiently small, summing (80) over all $(r, k) \neq (1, 1)$ gives $O(q)$ uniformly in m . This follows from the convergent double geometric series in $qe^{2C_0 q}$; its smallest remaining exponent $rk - 1$ is one. Explicitly, writing $t = qe^{2C_0 q}$ and enlarging the sum to all positive integers r , one may use

$$\sum_{k, r \geq 1} rk^2 t^{rk} = \sum_{k \geq 1} \frac{k^2 t^k}{(1 - t^k)^2} \leq \frac{1}{(1 - t)^2} \sum_{k \geq 1} k^2 t^k.$$

After removing the leading term, this analytic majorant is $O(t^2)$, which becomes $O(q)$ after division by q .

Consider the projected iteration

$$\mathcal{T}_q(\delta) = \delta - \frac{1}{4} P \nabla F_q(\delta). \quad (81)$$

It preserves the zero-sum subspace. At zero, $\mathcal{T}_q(0) = -Pg/(8q) = O_\infty(q)$ by [theorem 8.1](#). On a ball $\|\delta\|_\infty \leq C_0q$, its derivative on the zero-sum subspace has sup operator norm $O(q)$: the derivative of the leading term differs from the identity by $O(q)$, and the remaining Hessian has the bound just proved. Choose C_0 larger than twice the absolute constant in $\|\mathcal{T}_q(0)\|_\infty \leq Cq$; then choose q_0 so that this derivative norm is at most $1/2$. The map is a contraction of the ball into itself. It therefore has a unique fixed point, which is a constrained stationary point of E_n . For small q , $h = -\frac{1}{2} \log q$ and $h + \delta_i > 0$, so the point lies in the positive-gap simplex. Strict convexity in [theorem 2.2](#) makes it the global minimizer.

For the unnormalized Hessian, the nearest-neighbor part is $\text{diag}(V''(h + \delta_i)) = 8qI + O(q^2)$ on this ball. The remaining symmetric Hessian has row sums $O(q^2)$ by the same estimate, multiplied by $2q$. A symmetric matrix's Euclidean operator norm is bounded by its maximum absolute row sum. This proves (79).

It remains to obtain the Euclidean, rather than only sup-norm, estimate in (78). Projected stationarity and integration along the segment give

$$0 = Pg + P \left(\int_0^1 \nabla^2 E_n(h + t\delta) dt \right) \delta.$$

The averaged Hessian restricted to the zero-sum subspace is bounded below by $4qI$ after decreasing q_0 . Taking its inner product with δ and applying [theorem 8.1](#) yields $4q \|\delta\|_2^2 \leq \|Pg\|_2 \|\delta\|_2$, proving $\|\delta\|_2 = O(q)$. $\square$

Remark 8.3 (Why uniformity is not automatic). A finite-dimensional implicit-function theorem, applied separately for each m , would only produce a radius and constants that might depend on m . The window row-sum estimate (80), the exponentially localized grid force, and the projected contraction are the ingredients that remove that dependence. Both sup and Euclidean norms are needed: the first locates the true minimizer, and the second prevents its energy error from acquiring an unwanted factor of m .

8.3 The boundary theorem

Theorem 8.4 (Uniform dilute gap and energy corrections). *There are absolute constants $q_0, C > 0$ such that for every $n \geq 3$ and $0 < q = e^{-2h} < q_0$, the optimal gaps on $[0, (n-1)h]$ obey*

$$\delta = -\frac{q}{2}Pe + O_{\ell^2}(q^2), \quad (82)$$

and

$$\left| E_n^*((n-1)h) - U_n(h) + 2\frac{n-3}{n-1}q^3 \right| \leq Cq^4. \quad (83)$$

In particular, the two endpoint gaps have leading displacement $-q/2 + q/(n-1)$, while every other gap has leading displacement $q/(n-1)$. The error in this whole vector is bounded in Euclidean norm by Cq^2 , independently of n .

Proof. Let $p = Pg$ and let A be the averaged projected Hessian on the zero-sum subspace between the grid and the optimizer. By [theorem 8.2](#), $A = 8qI + O(q^2)$, and stationarity says $A\delta = -p$. Thus

$$\delta = -\frac{p}{8q} + O_{\ell^2}(q^2) = -\frac{q}{2}Pe + O_{\ell^2}(q^2),$$

where the last equality uses [theorem 8.1](#).

Taylor's theorem, (79), and $\|\delta\|_2 = O(q)$ give

$$E_n^* - U_n = p \cdot \delta + 4q \|\delta\|_2^2 + O(q^4) = -\frac{\|p\|_2^2}{16q} + O(q^4).$$

The projected grid-force expansion has an $O(q^3)$ Euclidean remainder, so

$$\|p\|_2^2 = 16q^4 \|Pe\|_2^2 + O(q^5).$$

Finally,

$$\|Pe\|_2^2 = \|e\|_2^2 - \frac{(\mathbf{1}^\top e)^2}{m} = 2 - \frac{4}{m} = 2\frac{n-3}{n-1}.$$

Substitution proves (83). Every estimate used an absolute constant, so the asserted uniformity follows. $\square$

For $n = 3$, $Pe = 0$ and symmetry makes the grid exactly optimal. For $n = 2$ the grid is again exact but is excluded from the vector notation above. For $n \geq 4$ the leading correction is strictly negative in the optimized energy: compressing the edge gaps and slightly enlarging the interior gaps beats the grid. The gain has order e^{-6h} even though the leading interaction has order e^{-2h} .

9 A sharp determinant transition and three correction orders

9.1 The leading transition from finite inequalities

Proposition 9.1 (The dilute trichotomy). *Uniformly in $n \geq 2$, as $h \rightarrow \infty$,*

$$E_n^*((n-1)h) = (n-1)V(h)(1 + O(e^{-2h})). \quad (84)$$

If $h_n \rightarrow \infty$ and $ne^{-2h_n} \rightarrow a \in [0, \infty]$, then

$$\mathcal{D}_n((n-1)h_n) \longrightarrow \begin{cases} 1, & a = 0, \\ e^{-4a}, & 0 < a < \infty, \\ 0, & a = \infty. \end{cases} \quad (85)$$

Proof. The nearest-neighbor Jensen bound gives $E_n^* \geq (n-1)V(h)$. On the grid,

$$U_n(h) \leq (n-1) \sum_{k \geq 1} V(kh) = (n-1)(V(h) + O(q^2)), \quad q = e^{-2h}.$$

Since $V(h) = 2q + O(q^3)$, the relative error is $O(q)$ uniformly in n . This proves (84) and then (85). $\square$

The transition interval length is therefore of order $\frac{1}{2}n \log n$, not of order n . Its profile resembles an extreme-value expression, but the problem here is entirely deterministic; no random extremal limit law is being claimed.

9.2 Where the boundary correction first becomes visible

Theorem 9.2 (Three-order critical-window expansion). *Let s range over a fixed compact subset of $\mathbb{R}$, put*

$$h_n = \frac{1}{2} \log n + s, \quad a = e^{-2s}, \quad L_n = (n-1)h_n. \quad (86)$$

Then, uniformly in that compact set,

$$\begin{aligned} E_n^*(L_n) &= 2a + \frac{2a^2 - 2a}{n} + \frac{\frac{8}{3}a^3 - 4a^2}{n^2} \\ &\quad + \frac{2a^4 - \frac{26}{3}a^3}{n^3} + O(n^{-4}). \end{aligned} \quad (87)$$

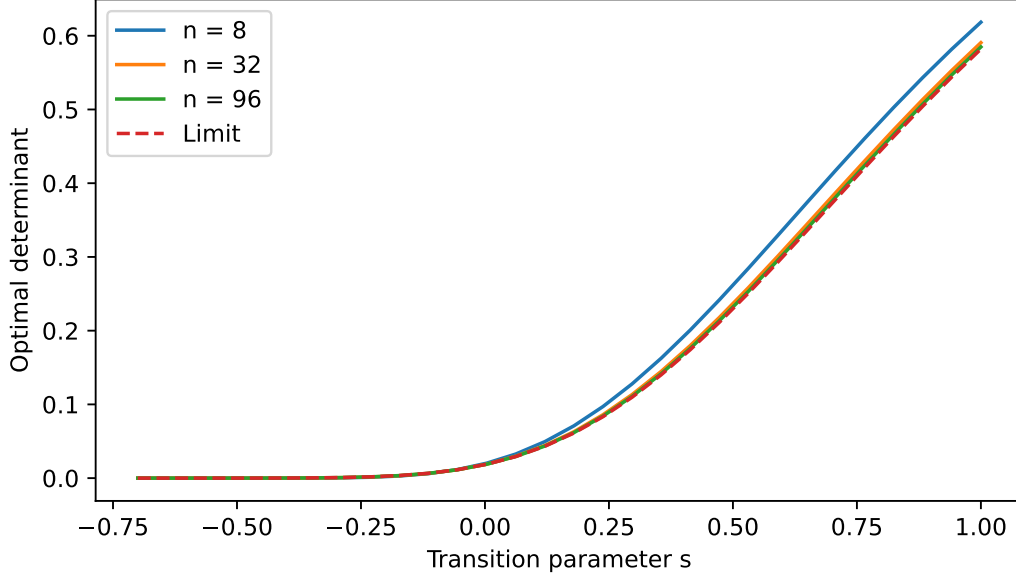


Figure 2: Numerically optimized determinants at $L = (n - 1)(\frac{1}{2} \log n + s)$ compared with the proved limit $\exp(-4e^{-2s})$. Curves are numerical diagnostics; the proof of convergence is independent of the optimization code.

Consequently

$$\begin{aligned} \log \mathcal{D}_n(L_n) = & -4a - \frac{4a^2 - 4a}{n} - \frac{\frac{16}{3}a^3 - 8a^2}{n^2} \\ & - \frac{4a^4 - \frac{52}{3}a^3}{n^3} + O(n^{-4}). \end{aligned} \quad (88)$$

The equal-grid energy has $-20a^3/3$, instead of $-26a^3/3$, in the cubic-in- a part of its n^{-3} coefficient.

Proof. For $n \geq 5$, expand the grid energy by shells using $V(kh) = 2q^k + \frac{2}{3}q^{3k} + \frac{2}{5}q^{5k} + \dots$. Through degree four,

$$U_n(h) = 2(n-1)q + 2(n-2)q^2 + \left(\frac{8}{3}n - \frac{20}{3}\right)q^3 + (2n-8)q^4 + O(nq^5). \quad (89)$$

The last error is uniform for q in a sufficiently small fixed interval: collecting the remaining positive terms and using $a_j^{\text{bulk}} \leq 2j$ bounds the tail by $2n \sum_{j \geq 5} jq^j = O(nq^5)$.

By [theorem 8.4](#), subtract $2(n-3)q^3/(n-1)$ from (89), with an additional absolute $O(q^4)$ error. Now insert $q = a/n$ and expand $(n-3)/(n-1) = 1 - 2/n + O(n^{-2})$. The two error bounds are $O(n^{-4})$ uniformly for bounded positive a , which holds on compact s -sets. Collecting powers of $1/n$ gives (87). Multiplication by -2 proves (88). $\square$

At $s = 0$, the n^{-1} term happens to vanish and the formula simplifies to

$$E_n^*\left(\frac{n-1}{2} \log n\right) = 2 - \frac{4}{3n^2} - \frac{20}{3n^3} + O(n^{-4}). \quad (90)$$

This cancellation is useful for testing the cubic correction numerically. It must not be interpreted as an n^{-2} leading error for every s .

10 Reproducible checks and numerical evidence

10.1 Verification protocol

The archive contains `verify.py`, its recorded output, machine-readable JSON and CSV tables, and the two figures. The program has no network dependency. The recorded environment was Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, mpmath 1.3.0, and SymPy 1.14.0. To reproduce the computations and regenerate the figures, run

```
python verify.py --output verification --figures
```

The program performs exact rational coefficient checks with SymPy, 80-decimal-digit quadrature and small-dimensional Newton calculations with mpmath, and double-precision convex optimization for larger designs. The latter uses the exact analytic gradient and Hessian with a line search; endpoints are fixed. No random samples enter these checks.

The checks address several independent parts of the argument: normalization and constant potential; the elliptic-series coefficients; inverse-capacity coefficients; eta modularity and the exact finite-product tail; the finite Jensen enclosure; the dilute energy and gap corrections; and the critical-window expansion. All programmed assertions passed in the recorded run. The mpmath results are high-precision numerical diagnostics, not rigorous enclosures, and the code is not a formal proof checker.

10.2 Equilibrium normalization and potential

The substitution $t^2 = 1 + (b^2 - 1)\sin^2 \theta$ removes the square-root endpoint singularities from normalization. For the potential, the integration interval is also split at its logarithmic singularity. At $L = 0.2, 1, 3$, normalization discrepancies were below 10^{-77} and potential discrepancies at $u/L = 0.1, 0.5, 0.9$ were below 10^{-79} at the working precision. A displayed zero in the raw output means zero after finite-precision rounding, not an exact symbolic integration certificate. The equilibrium energies are

L	I_L
0.2	2.99904551559535685251
1	1.45927489567825624541
3	0.66810270286063430161

For comparison, the fixed-interval minimizing energies at $L = 1$ are:

n	$2E_n^*(1)/(n(n-1))$ (numerical)
8	1.070740746589
16	1.232683639463
32	1.327711901636
64	1.383689948640
128	1.416371676477

The convergence is visibly slower than the dilute critical-window convergence. The table does not supply a rate for [theorem 4.1](#), and it would be inappropriate to infer the first edge gap from these energy values.

10.3 A direct test of the boundary energy gain

Define the scaled grid-to-optimum gain

$$G_{n,h} = e^{6h}(U_n(h) - E_n^*((n-1)h)). \quad (91)$$

The theorem predicts $G_{n,h} \rightarrow 2(n-3)/(n-1)$ for each n , with an error bounded by an absolute constant times e^{-2h} uniformly in n . The 80-digit calculations give:

n	$G_{n,3}$	$G_{n,4}$	Predicted limit
4	0.666287327830	0.666616746258	0.666666666667
8	1.435241666559	1.429474991882	1.428571428571
16	1.743372444134	1.734691207867	1.733333333333

The raw file also records the energy error divided by q^4 and the Euclidean gap error divided by q^2 . Those quantities stay of ordinary size in the tested cases, as predicted. Finite tests cannot establish uniformity over all dimensions; that is supplied by the proof in [section 8](#).

10.4 The critical window

At $s = 0$, let $E_n^{(3)} = 2 - 4/(3n^2) - 20/(3n^3)$. The following values test the three-order approximation:

n	Optimized E_n	$\mathcal{D}_n$	$n^4(E_n - E_n^{(3)})$
8	1.964989270450	0.019644094093	-4.73728
16	1.993072791619	0.018571157326	-5.98153
32	1.998488269347	0.018371099313	-6.49781
64	1.999648646953	0.018328513923	-6.72596
128	1.999915415425	0.018318737592	-6.83227

The limiting determinant is $e^{-4} = 0.018315638887\dots$. The last column is consistent with an $O(n^{-4})$ error but is not used to guess or assert the next exact coefficient. The recorded stationarity residuals for these critical-window optimizations range from about 3.1×10^{-13} to 3.6×10^{-15} .

10.5 Product and inverse checks

The direct and modular bulk formulas were compared at $h = 0.2, 0.5, 1, 2$. Their discrepancies were below 10^{-79} at 80-digit precision. The identity $U_n = nS - T + R_n$ was checked at the same values with $n = 9$, also below 10^{-78} . Exact symbolic calculations reproduce the first five coefficients of $B(z)$, the first five coefficients in (46), and the full polynomial in (87). An additional inverse check evaluates (46) at the exact elliptic energies for $L = 1, 2, 3$; its residual divided by Q^6 approaches the next coefficient -4 , which is obtained from (45) rather than fitted from the data.

11 What remains open and a proposed research program

The following questions are deliberately separated from the proved results. They range from concrete extensions of the present estimates to more structural transseries and formalization projects.

11.1 Microscopic fixed-interval endpoint spacings

Research question 11.1. For fixed $L > 0$ and fixed $j \geq 1$, determine the limit, if it exists, of $n^2 u_{j+1}^{(n)}$, with $u_1^{(n)} = 0$. Obtain the reflected right-edge law and a transition to the bulk spacing scale.

The density coefficient in (30) identifies the natural edge scale but not the discrete constant. A promising approach is to rewrite (18) using $t_i = e^{2u_i}$, derive an equation for the node polynomial or its logarithmic derivative, and study the endpoint boundary layer. A proof would complete the endpoint portion of `question:design-limit`; that portion is not settled here.

11.2 The finite-density surface energy

Research question 11.2. For each fixed $h > 0$, does

$$\Sigma(h) = \lim_{n \rightarrow \infty} (E_n^*((n-1)h) - nS(h))$$

exist? Can it be represented by a uniquely minimizing semi-infinite boundary profile and an explicit correction for the fixed total length?

The bounds in [theorem 6.1](#) place the expression in a compact interval, but boundedness alone does not prove a limit. The equal-grid contribution is $-T(h)$. Any additional term measures boundary relaxation. An existence proof should control how the two edge layers communicate through the length constraint, not merely invoke exponential decay of pair forces.

11.3 Higher uniform dilute coefficients

Research question 11.3. Develop, for every fixed N , a dimension-uniform expansion of the optimal gap vector and energy through q^N . Determine which coefficients stabilize as $n \rightarrow \infty$, and isolate the rational $1/(n-1)$ corrections caused by the zero-sum projection.

The uniformly contractive representation in [theorem 8.2](#) provides a starting point. The challenge is to preserve an absolute energy remainder rather than an n -dependent bound at each order. The q^4 surface coefficient would immediately determine the n^{-4} critical-window coefficient. The numerical table is useful as a regression test, but coefficient identities should be derived symbolically.

11.4 A mesoscopic bridge between elliptic and crystalline limits

Research question 11.4. Find uniform asymptotics when $L = L_n \rightarrow \infty$ but $h_n = L_n/(n-1) \rightarrow 0$. Which subregimes permit the fixed-interval elliptic energy to match the small- h expansion of $nS(h)$, and where do entropy-like discrete corrections first change the answer?

The fixed- L theorem and the positive-gap theorem cannot be interchanged without proof. Formally, $I_L \sim \pi^2/(4L)$ and $S(h) \sim \pi^2/(8h)$ yield matching leading energies $\pi^2 n^2/(8L)$, but the next orders contain different boundary and discreteness information. A uniform comparison would connect the elliptic and modular transseries in a genuinely two-parameter theorem.

11.5 The full hyperbolic superfactorial crossover

Research question 11.5. Starting from (65), determine a complete joint asymptotic as $h \downarrow 0$ and $n \rightarrow \infty$, uniformly in prescribed ranges of nh . Include the MacMahon surface contribution and the exact finite-size tail at the same precision.

The exact factorization solves an algebraic part of the repository's `question:q-barnes`, but not its full asymptotic request. A Mellin representation of $\log M(e^{-t})$ is a natural route to power-logarithmic terms; a further analysis is needed to justify optimal truncation and exponentially small corrections. The convergent eta expansion for S is not a substitute for that analysis.

11.6 General screened logarithmic interactions

Research question 11.6. Suppose W is decreasing, strictly convex, logarithmically singular at zero, and has a convergent large-distance expansion $W(x) = c_1 e^{-\alpha x} + c_2 e^{-2\alpha x} + \dots$, with $c_1 > 0$. Which hypotheses give a dimension-uniform boundary correction analogous to [theorem 8.4](#), and which coefficients determine its first term?

The Jensen certificates and bounded-defect argument use much less than the special hyperbolic form. The projected grid force and Hessian method suggest a general theorem for sufficiently controlled exponential tails. One should distinguish the universal boundary mechanism from the exact elliptic equilibrium and determinant identities, which are special here.

11.7 Several allowed intervals and external fields

Research question 11.7. Determine equilibrium measures when the admissible u -set is a union of intervals, or when the objective includes a smooth external field. Can one classify support transitions and produce uniform expansions near a newly opening or closing support component?

Under $t = e^{2u}$, the one-interval calculation becomes an arcsine Cauchy integral. Several intervals lead to a more complicated algebraic singular-integral problem with unknown filling fractions. This offers a concrete setting for multi-scale inversion and critical transseries, but no multi-interval formula is claimed in the present paper.

11.8 Covariance spectrum beyond the determinant

Research question 11.8. At the optimal configurations, determine the smallest eigenvalue, condition number, and empirical eigenvalue distribution of $[\text{sech}(u_i - u_j)]$. How do they behave in the fixed- L , positive-gap, and dilute critical regimes?

A determinant is only the product of eigenvalues. The limit [\(10\)](#) alone does not yield a lower bound on every eigenvalue or establish numerical stability of interpolation. The Cauchy-matrix representation may permit explicit inverse formulas and more refined spectral estimates.

11.9 Sharp nonasymptotic numerical certification

Research question 11.9. Combine interval arithmetic, convexity, and the explicit Hessian with [\(50\)](#) to produce rigorous finite- n enclosures of the true optimizer, its determinant, and its edge gaps. Can the certified cost be reduced below dense cubic-time Newton solves?

The supplied code is a transparent diagnostic, not a complexity-optimized solver. The covariance Cauchy structure and exponentially decaying interactions suggest structured linear algebra or certified truncation, with distinct choices in dense and dilute regimes.

11.10 Formal proof of the finite-dimensional core

Research question 11.10. Formalize the determinant identity, Jensen enclosure, and projected boundary estimates in Lean, and then connect them to a formalized measure-theoretic equilibrium theorem.

The finite inequalities are natural first milestones: they avoid principal-value integration and elliptic special functions while already providing useful certified determinant estimates. A later stage should formalize the arcsine integral and its endpoint continuity, rather than introducing the whole equilibrium theorem as an axiom.

12 Formalization boundary and proof dependency map

No Lean code is included or claimed to have compiled. The following partition identifies what a future formalization would actually need.

Finite algebra and inequalities. The Cauchy determinant identity, geometric series, derivatives of V , strict convexity, finite Jensen bounds, and the count $N_{i,k}$ are the most self-contained layer. The main dependencies are finite sums, matrices, elementary real analysis, and constrained convex optimization. The dimension-free window bounds require estimates uniform in a natural number parameter, not an infinite-dimensional optimizer.

Analytic fixed-point layer. The projected contraction and its sup/Euclidean norm conversion establish [theorem 8.4](#). A formal proof should expose the constants and a permitted q_0 , instead of hiding them in successive uses of $O(q)$. The manuscript's existence of absolute constants follows from explicit convergent majorant sums; it does not yet provide a machine-readable optimized numerical threshold.

Measure and special-function layer. The continuum proof requires logarithmic-potential continuity, principal-value differentiation, a Fourier positivity argument for finite signed measures, and weak compactness. The elliptic definitions are ordinary real integrals. The capacity transseries then adds the standard elliptic expansion, while the bulk modular transseries adds the eta transformation. These are logically separate inputs.

Dependency summary. The chain for the new finite-size result is

$$\begin{aligned} \text{window counting} &\implies \text{localized grid force} \implies \text{uniform projected contraction} \\ &\implies \text{Euclidean Hessian control} \implies \text{boundary energy correction} \\ &\implies \text{critical-window expansion.} \end{aligned}$$

It does not depend on the continuum equilibrium theorem or on elliptic or modular special-function identities. Conversely, the fixed- L equilibrium theorem is independent of the dilute expansion. This separation gives independent audit targets and prevents a numerical check of one regime from being misrepresented as evidence for another.

13 Conclusion

The fixed-interval design question from the Fabius report has an explicit answer at the level of equilibrium measure and optimal leading energy. Its density is [\(6\)](#), its constant potential is [\(7\)](#), and its edge enhancement and screened bulk are described by [theorems 4.2 and 4.3](#). These formulas connect the repository to classical elliptic Green capacity in a precise normalization. Their expansion and inversion supply convergent, quantitatively controlled examples for the transseries program.

The finite-design problem has additional structure that the equilibrium measure does not capture. The finite Jensen enclosure works for every n, L ; the positive-gap limit has bounded total gap defect; and the dilute boundary correction is an absolute, dimension-uniform expansion. That last point makes it possible to resolve the deterministic determinant transition and calculate three finite-size correction orders without replacing the optimum by an unjustified uniform-grid approximation.

The remaining questions are substantive. In particular, the fixed-interval microscopic edge problem, the finite-density surface energy, and the full joint MacMahon crossover are not hidden inside the results proved here. They are explicit next targets with concrete formulas and verification infrastructure on which to build.

A Source provenance and reproducibility

The inspected repository is VladimirReshetnikov/ProveIt. The source report's path is

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/common_digit_fabius_zonoids_frontier_report/common_digit_fabius_zonoids.tex.
```

The repository search identified commit 0c973d8f5e7ef4550f4df1bf486d890037fd9f14; a direct read at that commit confirmed the question block at source lines 1744–1785. The blob identifier is dabfc501dfe952d826001703a087b47424568581. The finite node-design statement was inspected in the same source's 1370–1590 range. The exact label of the primary question is `question:design-limit`; the adjacent product question is `question:q-barnes`. The source was accessed on 29 September 2026.

The transseries collection's README was also inspected to understand the repository's current organization and its distinction between formalized modules and unreviewed research arrivals. It was not used as evidence that any theorem in this manuscript is formalized. No repository files were modified during this work.

The archive contains the standalone source `article.tex`, the compiled `article.pdf`, `verify.py`, the recorded `verification.log`, tables and figure files under `verification/`, and a short README and provenance file. The article can be rebuilt with

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied figure PDFs are sufficient for rebuilding; regenerating them requires Matplotlib. Font files and third-party research PDFs are not redistributed in the package.

B A convenient coefficient ledger

For exact regression testing, several derived quantities are collected here with their independent definitions.

j	$[z^j]B(z)$	$[Q^j](L - X + \log 2)$	c_j in (67)
1	1/4	2	−1
2	13/128	−3	1/2
3	23/384	8/3	−4/3
4	2701/65536	−3/2	5/4
5	5057/163840	12/5	−6/5

The two occurrences of Q in this ledger belong to different problems: for inverse capacity, $Q = e^{-\pi^2/I}$; for the bulk energy, $Q = e^{-\pi^2/h}$. They are not being identified. The first column is obtained from $B = D/A$, the second from (45), and the third from divisor sums. All entries are exact rational numbers.

The critical-window energy coefficients, in powers of $1/n$, are

$$\begin{aligned} C_0(a) &= 2a, \\ C_1(a) &= 2a^2 - 2a, \\ C_2(a) &= \frac{8}{3}a^3 - 4a^2, \\ C_3(a) &= 2a^4 - \frac{26}{3}a^3. \end{aligned}$$

In particular, the boundary relaxation contributes exactly $-2a^3$ to C_3 , and no contribution to C_0, C_1, C_2 . This is a useful normalization check: accidentally using the product energy in place of the log determinant would change every final logarithmic coefficient by a factor of two.

References

- [1] VladimirReshetnikov/ProveIt repository, *Common-Digit Fabius Zonoids: Exact volumes, hyperbolic-secant geometry, Bernoulli Gaussianization, and parameter jets*, research report, commit 0c973d8f5e7ef4550f4df1bf486d890037fd9f14. The source path and question labels are recorded in [section A](#). Accessed 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
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- [5] NIST Digital Library of Mathematical Functions, Chapter 23, §23.18, *Modular Transformations*, particularly the Dedekind eta transformation, formula 23.18.5. Accessed 29 September 2026. <https://dlmf.nist.gov/23.18>.